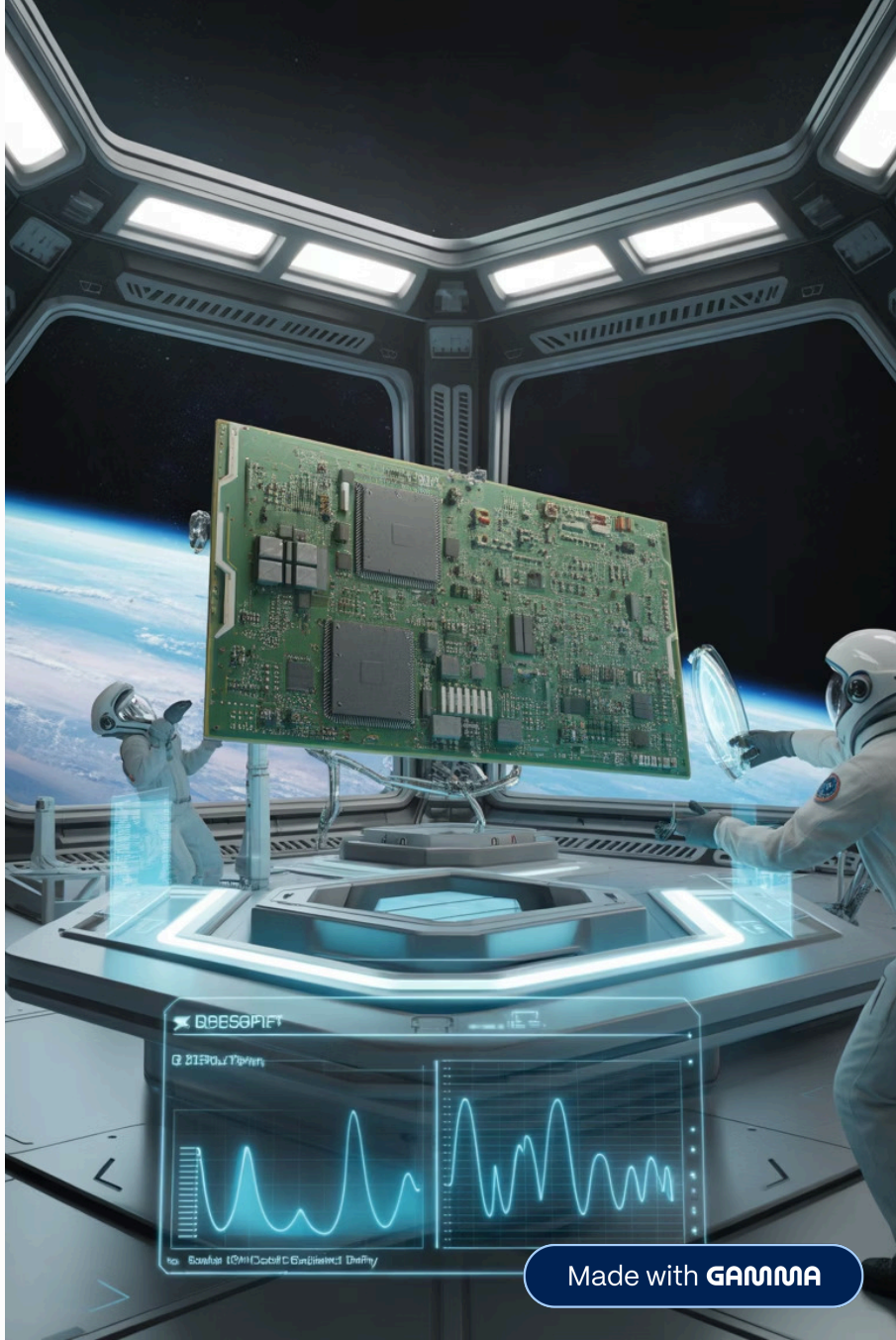
# PCB Deformation & Scale Coefficient Drift

An investigation into thermomechanical failure modes causing measurable calibration drift in spacecraft modules following environmental qualification testing.

ACTIVE INVESTIGATION

TVC / TVCH POST-TESTING



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Problem Overview

# A Persistent Calibration Anomaly

## Observed Symptom

PCB deformation detected within the module, correlated with a **3–5% scale coefficient drift** in affected units.

## Trigger Condition

Deviation consistently appears **after TVC or TVCh environmental testing**, then remains stable throughout spacecraft operation.

## Current Mitigation

No root cause identified. The operational response is **full module replacement**, which restores performance but resolves nothing.

## Open Question

A tolerance expansion to **accept the 3–5% drift** is under discussion — a proposal that carries significant technical risk.

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# Confirmed Observed Facts

## PCB deformation detected

Physical deformation confirmed on inspected units following environmental qualification.

## 3–5% scale coefficient drift measured

Drift is consistent across affected units; magnitude falls within a narrow, repeatable band.

## Drift is stable post-testing

Once induced, the deviation does not evolve further during nominal spacecraft operation.

## Root cause remains unknown

No causal mechanism has been confirmed. Investigation is ongoing.

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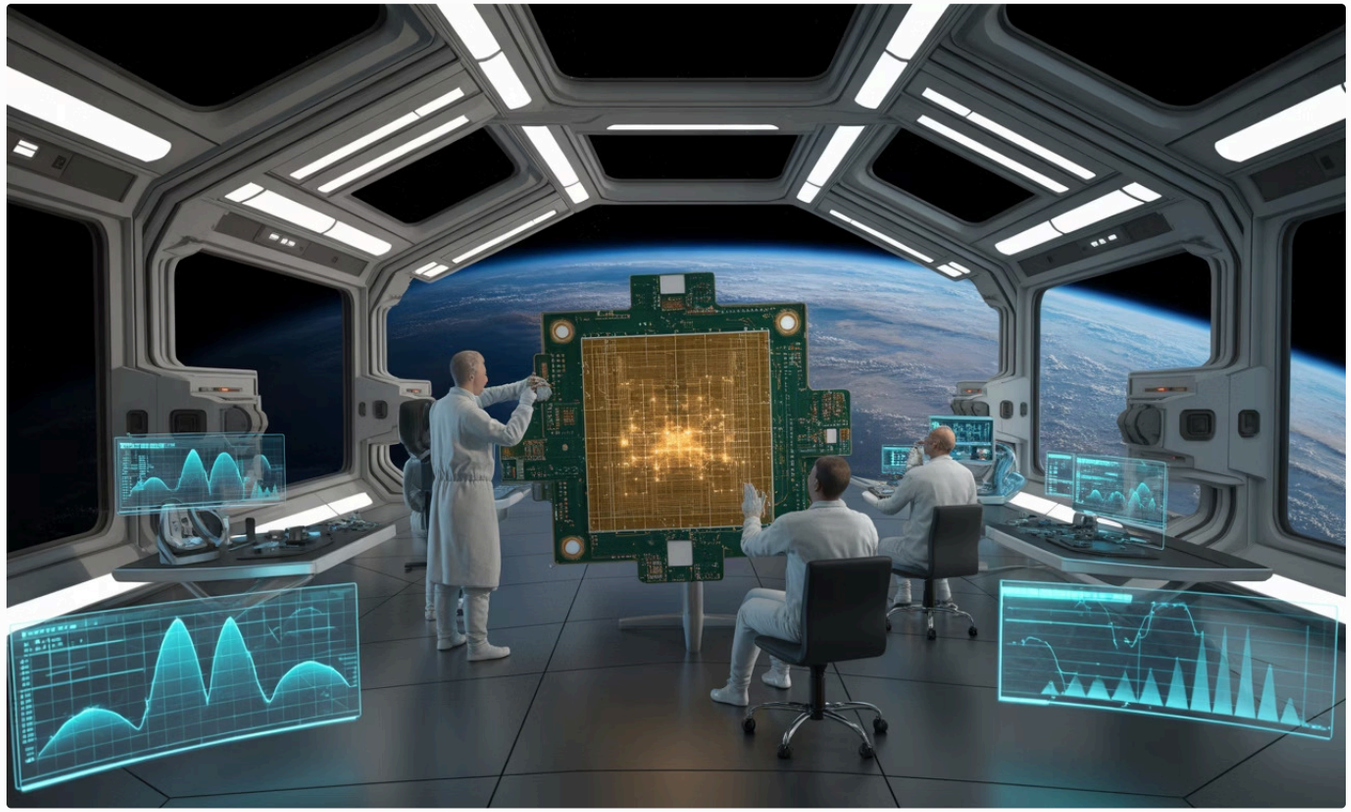
Preliminary Interpretation

# A Permanent Structural Change During Testing



The pattern — drift appearing after environmental testing and remaining stable — is consistent with a **permanent thermomechanical state change** induced during the test cycle.

The physical structure of the board or its mounting changes irreversibly. The electronics respond to this new mechanical baseline, producing a measurable shift in calibrated electrical parameters.

### Mechanical

Residual stress, warpage, geometry shift

### Electrical

Analog drift, solder fatigue, calibration offset

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Hypotheses

# Five Candidate Failure Mechanisms

1

### Thermomechanical PCB Deformation

Residual stress from TVC induces bending, warping near sensitive components, or shifts in mechanical reference geometry.

2

### CTE Mismatch

Differential thermal expansion between substrate, frame, adhesives, and sensor packages accumulates plastic deformation across cycles.

3

### Mounting Constraints

Rigid fixturing without thermal compensation may permanently warp the board under repeated thermal load.

4

### Analog Component Drift

Resistor networks, reference voltages, or amplifier parameters may shift due to environmental stress — independent of mechanical effects.

5

### Solder Joint Micro-Fractures

Thermal fatigue creates micro-cracks in joints or vias, altering signal paths and introducing measurable calibration drift without causing outright failure.

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Risk Assessment

# Tolerance Expansion: A High-Risk Workaround

## Potential Benefits

- Fewer rejected units at acceptance
- Simplified qualification criteria
- Reduced module replacement rate

## Risks Introduced

- Drift mechanism may continue to evolve in-orbit
- May mask correlated hidden degradation
- Production instability remains undetected
- Compromises long-term mission reliability

**Assessment:** Accepting an unexplained parameter deviation in a spacecraft system — without identifying its mechanism — introduces unquantifiable risk. The apparent 3–5% margin relative to the nominal 9% threshold narrows the safety band to a level that cannot be reliably verified without understanding the underlying cause.

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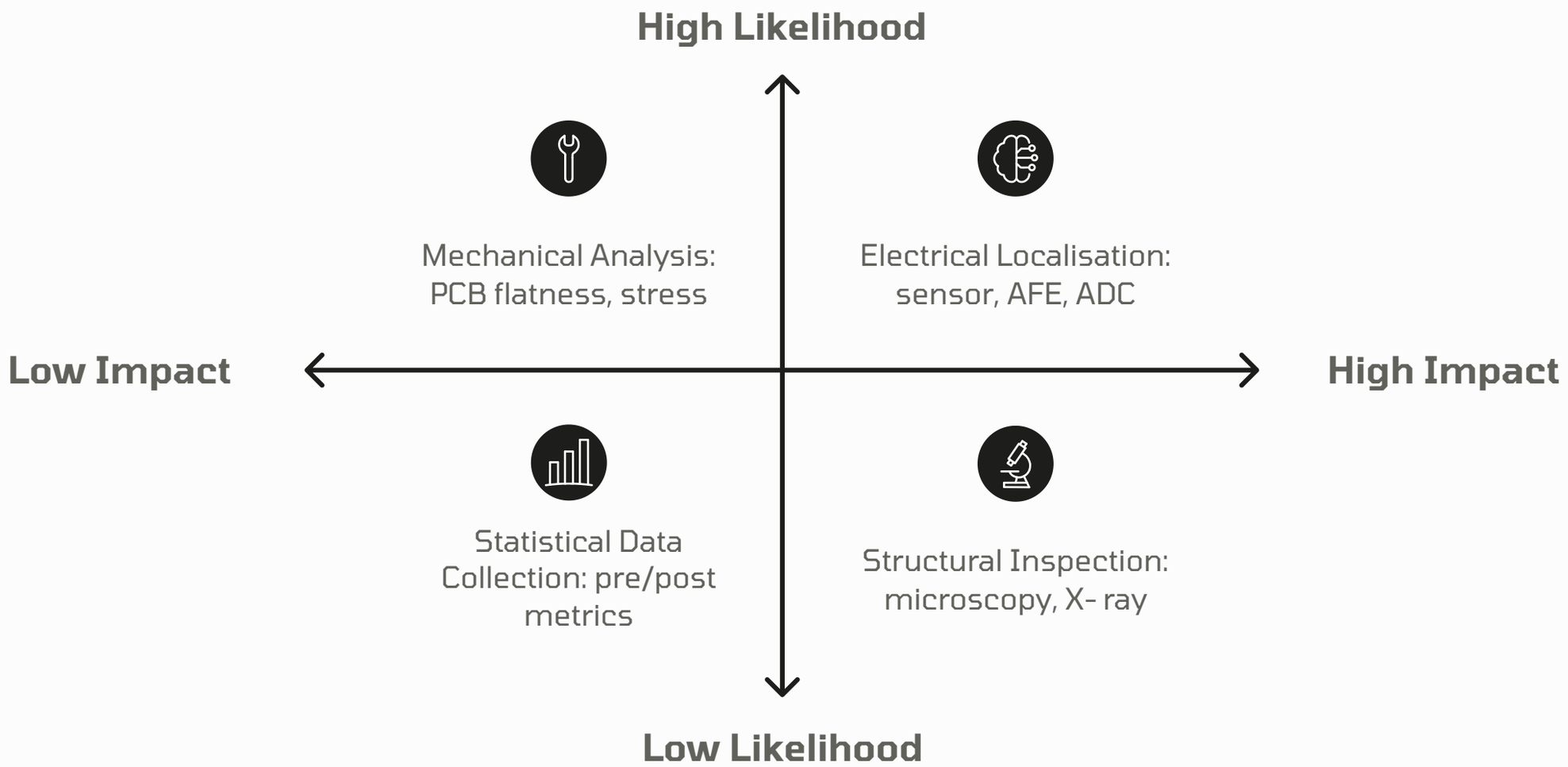
# Required Investigation: Four Workstreams




Each workstream targets a distinct layer of the failure hypothesis space. Parallel execution minimises time to root cause identification.

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Workstream Detail

# Mechanical & Electrical Investigation Priorities

## Mechanical Analysis

- Measure PCB flatness before and after each test phase
- Quantify deformation magnitude and spatial distribution
- Analyse stress at mounting points; evaluate fastener torque effects
- Map CTE mismatch between substrate, frame, and potting

## Electrical Parameter Localisation

Isolate which circuit block generates the drift:

01

**Sensor element**

02

**Analogue front end**

03

**Reference voltage circuit**

04

**ADC calibration stage**

05

**Power supply stability**

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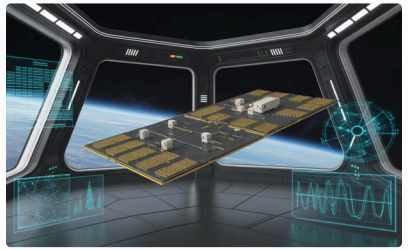
Structural Inspection Protocol

# Defective Units Require Detailed Teardown

## Solder Joint Microscopy

Inspect for micro-fractures, cold joints, and fatigue cracks at high magnification across all critical interconnects.



## X-Ray Inspection

Examine hidden joints, via integrity, and internal layer continuity — especially beneath BGA or QFN packages.

## Warpage Measurement

Quantify board bow and twist using profilometry; correlate spatial deformation with coefficient drift magnitude.

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# Conclusion & Recommended Path Forward

### Root cause unconfirmed

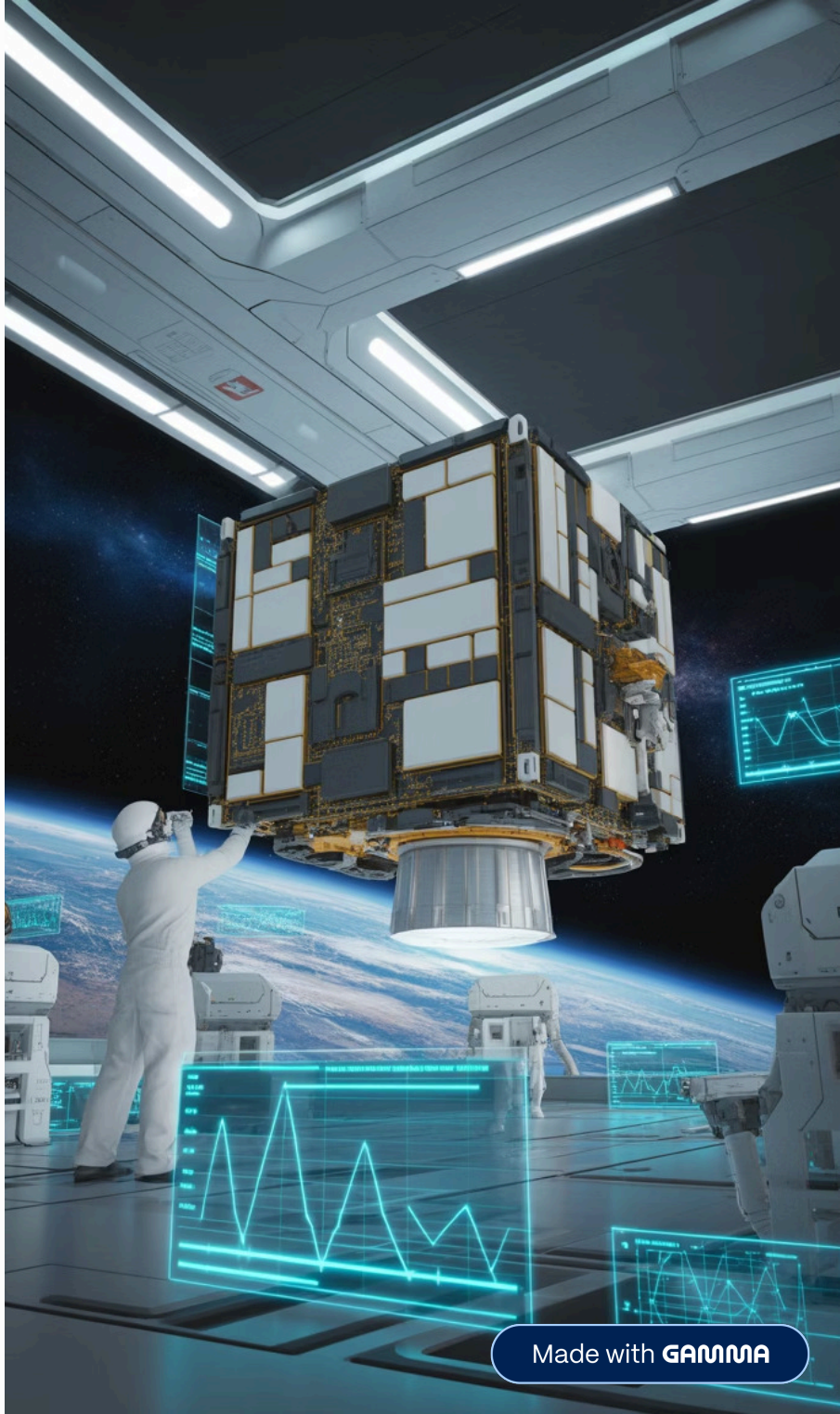
The 3–5% drift post-TVC/TVCh is reproducible, but no single causal mechanism has been verified. Thermomechanical and electrical hypotheses must both be tested.

### Do not expand tolerance prematurely

Accepting the deviation without causal understanding introduces unacceptable long-term risk to mission reliability and safety margins.

### Execute all four investigation workstreams

Mechanical analysis, electrical localisation, statistical data collection, and structural inspection must proceed in parallel to close the knowledge gap efficiently.



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